

SameerTraderz ERP — Complete System Flow & Executive Briefing Guide

Executive Briefing & System Handbook

Created for Business Stakeholders, Operations Managers, Auditors, and System Administrators.

This guide explains the entire SameerTraderz ERP system from first principles, requiring zero prior technical or accounting background.

1. Executive Briefing (1-Page Quick Summary)

What is SameerTraderz ERP?

SameerTraderz ERP is an enterprise-grade wholesale distribution, inventory tracking, and double-entry financial accounting system built specifically for FMCG (Fast-Moving Consumer Goods) wholesale operations (such as hygiene products, tissues, soaps, and diapers).

What Does the System Do?

1. **Tracks Inventory in Real Time:** Tracks exact physical stock in single piece units using Weighted Average Costing (WAC). Every box bought or item sold immediately updates physical stock and financial stock value.

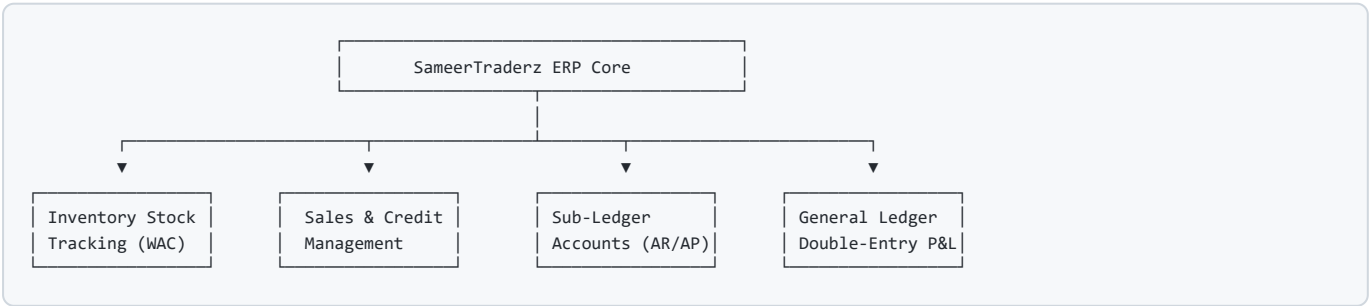
2. **Manages Wholesale Sales & Invoicing:** Handles cash sales, credit sales, salesman performance targets, discounts, transport allowances, and upfront cash collections.

3. **Automates Customer & Supplier Ledgers:** Tracks exact balances owed by customers (Accounts Receivable) and owed to suppliers (Accounts Payable).

4. **Enforces Double-Entry Accounting:** Every rupee moving through the business (sales, purchases, payments, returns, expenses) automatically posts balanced Debit and Credit entries to the General Ledger (GL).

5. **Protects Historical Financials (Period Locks):** Admin users can lock closed fiscal months. Once locked, no transactions or edits can alter historical financial records.

2. Core Business Architecture



3. The 6 Complete Business Lifecycle Flows

FLOW 1: Stock Purchasing & Procurement (Vendor Buying)

flowchart LR

```
A["Supplier Order"] --> B["Record Purchase"]
B --> C["Stock Increment (+Pieces)"]
B --> D["Recalculate Weighted Avg Cost (WAC)"]
B --> E["Update Supplier Ledger"]
B --> F["Post GL Journal Entry"]
```

What Happens Step-by-Step:

1. **Purchase Entry:** Operator records a purchase from a Supplier specifying items, piece quantities, and purchase prices.
2. **Stock Increment:** Physical stock quantity (`Product.stockQuantity`) is automatically increased in piece units.
3. **Weighted Average Cost (WAC) Update:** The system re-calculates the product's average unit cost:
$$\text{New WAC} = \frac{(\text{Existing Stock} \times \text{Existing WAC}) + (\text{New Qty} \times \text{New Unit Cost})}{\text{Total Stock Quantity}}$$
4. **Supplier Ledger Record:**
 - If unpaid: Increases what we owe the supplier (Credit `SupplierLedger`).
 - If partial cash paid: Records payment debit to supplier.
5. **General Ledger (GL) Posting:**
 - **Debit:** `1300 - Inventory Asset` (Value of goods received)
 - **Credit:** `1000 - Cash in Hand` (Cash paid upfront)
 - **Credit:** `2000 - Accounts Payable` (Remaining amount owed)

FLOW 2: Sales Invoicing & Revenue Generation (Selling)

flowchart LR

```
A["Customer Order"] --> B["Create Invoice (Cash / Credit)"]
B --> C["Deduct Stock (-Pieces)"]
B --> D["Freeze COGS Fact"]
B --> E["Update Customer Ledger"]
B --> F["Post GL Revenue & COGS"]
```

What Happens Step-by-Step:

1. **Invoice Creation:** Sales representative or counter staff creates a sales invoice for a customer (Cash sale or Credit sale).
 2. **Stock Deduction:** Physical stock quantity (`Product.stockQuantity`) is instantly deducted.
 3. **Freezing Historical Cost of Goods Sold (COGS):**
 - The current unit WAC at the moment of sale is **frozen forever** into `InvoiceItem.costPriceAtSale` .
 - Changing future stock costs will **never** alter the profit calculated for this sale.
 4. **Customer Ledger Record:**
 - For Credit Sales: Debits `CustomerLedger` (increases what customer owes us).
 - If upfront cash collected: Credits `CustomerLedger` for cash received.
 - If store credit applied: Consumes customer's store credit.
 5. **General Ledger (GL) Posting:**
 - **Debit:** `1000 - Cash in Hand` (Cash collected)
 - **Debit:** `1100 - Accounts Receivable` (Unpaid credit invoice balance)
 - **Debit:** `2100 - Customer Deposits` (Store credit applied)
 - **Credit:** `4000 - Sales Revenue` (Gross invoice sales total)
 - **Debit:** `5000 - Cost of Goods Sold` (Total frozen COGS)
 - **Credit:** `1300 - Inventory Asset` (Inventory reduction value)
-

FLOW 3: Customer Sales Returns & Store Credit

```
flowchart LR
  A["Returned Goods"] --> B["Create Sales Return"]
  B --> C["Restock Inventory (+Pieces)"]
  B --> D["Issue Store Credit / Cash Refund"]
  B --> E["Reverse GL COGS & Revenue"]
```

What Happens Step-by-Step:

1. **Sales Return Processing:** Operator creates a sales return against an existing invoice. The system enforces that return quantity cannot exceed remaining returnable items.
 2. **Inventory Restocked:** Returned items re-enter live physical stock at the current pool WAC.
 3. **Refund Selection:**
 - **Registered Customer:** Automatically credited as **Store Credit** (Customer balance becomes negative or credit balance increases).
 - **Walk-in / Cash Refund:** Cash refunded directly to customer.
 4. **General Ledger (GL) Posting:**
 - **Debit:** 4100 - Sales Returns & Allowances (Revenue deduction)
 - **Credit:** 1100 - Accounts Receivable OR 1000 - Cash in Hand
 - **Debit:** 1300 - Inventory Asset (Restocked inventory value)
 - **Credit:** 5000 - Cost of Goods Sold (COGS reversal)
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FLOW 4: Purchase Returns & Vendor Claims

flowchart LR

```
A["Return Goods to Supplier"] --> B["Create Purchase Return"]
B --> C["Deduct Stock (-Pieces)"]
B --> D["Reduce Vendor Debt / Cash Refund"]
B --> E["Post GL Variance Gain/Loss"]
```

What Happens Step-by-Step:

1. **Purchase Return Entry:** Operator processes a return of damaged or incorrect goods back to the supplier.
 2. **Stock Decrement:** Stock quantity is deducted at current pool WAC.
 3. **Vendor Settlement:**
 - Reduces Accounts Payable liability owed to supplier (Debit **SupplierLedger**).
 - If cash refund received: Debits GL **1000** (Cash in Hand).
 4. **Variance Accounting:**
 - If agreed refund rate differs from stock pool WAC, the difference is posted to **5100 - Purchase Return Variance** as a Gain or Loss.
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FLOW 5: Customer & Supplier Payments & Allocations

flowchart LR

```
A["Payment Received / Made"] --> B["Record Payment"]
B --> C["Update Sub-Ledger Balance"]
B --> D["Allocate to Open Invoices / Bills"]
B --> E["Post GL Cash & AR/AP"]
```

What Happens Step-by-Step:

1. Customer Payment Received:

- Operator records customer cash/bank payment.
- Sub-ledger (`CustomerLedger`) credited $\rightarrow$ Customer balance decreases.
- GL Posting: Debit `1000 - Cash in Hand` , Credit `1100 - Accounts Receivable` .

2. Invoice Allocation: Payment amount is allocated against specific unpaid invoice balances (`balanceDue`).

3. Supplier Payment Made:

- Operator records payment to supplier.
- Sub-ledger (`SupplierLedger`) debited $\rightarrow$ Supplier balance decreases.
- GL Posting: Debit `2000 - Accounts Payable` , Credit `1000 - Cash in Hand` .

FLOW 6: Expenses & Fiscal Period Locks

flowchart LR

```
A["Business Expense"] --> B["Record Expense"] --> C["GL Debit 6000 Operating Expenses"]
D["End of Month / Fiscal Period"] --> E["Admin Locks Period"] --> F["Block All Retro-active Edits"]
```

What Happens Step-by-Step:

1. Business Expenses:

- Operating expenses (rent, electricity, salaries, tea/fuel) recorded with expense categories.
- GL Posting: Debit `6000 - Operating Expenses` , Credit `1000 - Cash in Hand` .
- Modifying or deleting expenses automatically cleans up or reverses GL entries.

2. Accounting Period Closing & Locking:

- Admin creates fiscal calendar periods (`July 2026` , `FY26 Q1`).
 - At month-end, Admin clicks **Close & Lock Period**.
 - Once locked, the `verifyPeriodNotClosed` guard intercepts all API attempts to create, edit, or delete transactions dated within that period, returning HTTP 400 rejection.
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4. Visual Financial & Inventory Summary Table

Business Action	Inventory Impact	Customer/Supplier Balance Impact	General Ledger (GL) Impact
Cash Sale	Stock Decreased (-Qty)	No Debt Change	Debit Cash (1000) / Credit Revenue (4000) Debit COGS (5000) / Credit Asset (1300)
Credit Sale	Stock Decreased (-Qty)	Customer Owes Us (+Debit)	Debit AR (1100) / Credit Revenue (4000) Debit COGS (5000) / Credit Asset (1300)
Credit Sale w/ Store Credit	Stock Decreased (-Qty)	Credit Balance Consumed	Debit Customer Deposits (2100) / Credit Revenue (4000)
Stock Purchase (Credit)	Stock Increased (+Qty) & WAC Recalculated	We Owe Supplier (+Credit)	Debit Asset (1300) / Credit AP (2000)
Customer Payment Received	No Stock Change	Customer Debt Reduced (-Credit)	Debit Cash (1000) / Credit AR (1100)
Supplier Payment Paid	No Stock Change	Supplier Debt Reduced (-Debit)	Debit AP (2000) / Credit Cash (1000)
Sales Return (Credit)	Stock Increased (+Qty)	Customer Debt Reduced (-Credit)	Debit Sales Return (4100) / Credit AR (1100) Debit Asset (1300) / Credit COGS (5000)
Purchase Return (Cash)	Stock Decreased (-Qty)	No Debt Change	Debit Cash (1000) / Credit Asset (1300) \$pm\$ Variance (5100)
Damaged Stock Adjustment	Stock Decreased (-Qty)	No Debt Change	Debit Inventory Shrinkage (5200) / Credit Asset (1300)
Operating Expense	No Stock Change	No Debt Change	Debit Expenses (6000) / Credit Cash (1000)

5. Module-by-Module Feature Reference

1. **Authentication & Security:** JWT Bearer Token authentication, role-based authorization (`ADMIN` vs `STAFF`), password hashing.
 2. **Master Data Management:** Customers, Suppliers, Products (SKU, barcode, unit price, WAC), Expense Categories, Salesmen.
 3. **Inventory Engine (`adjustStock`):** Enforces non-negative stock atomic SQL updates (`increment: delta`).
 4. **Sales & Invoicing Module:** Document state machine (`DRAFT` , `POSTED` , `VOIDED`), salesman target tracking, transport discounts.
 5. **Purchasing & WAC Engine:** Supplier purchases, automatic pool WAC recalculations.
 6. **Returns Module:** Sales returns with invoice balance settlement, purchase returns with variance accounting.
 7. **Payments & Store Credit Engine:** Prepayments, credit allocations, store credit applications, cash refunds.
 8. **General Ledger (GL) Engine:** 100% double-entry Trial Balance, Profit & Loss reporting, Balance Sheet accounts.
 9. **Accounting Period Locks:** Dynamic fiscal period creation, lock/unlock Admin controls, period closed error interceptors.
 10. **Reports & Executive Dashboard:** Real-time revenue, COGS, gross profit, net profit, low-stock alerts, sales by salesman performance.
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6. How to Brief This Software to Clients & Stakeholders

When presenting or briefing this system to business owners or managers, emphasize these 4 primary selling points:

1. **"Complete Financial Control":** Every rupee and every piece of stock is tied directly to a double-entry General Ledger. Money cannot disappear or be created without an audit trail.
2. **"Zero Historical Profit Tampering":** Unit costs are frozen at the exact moment of sale. Changing prices today will never alter last month's reported profits.
3. **"Period Locking Integrity":** Once an accounting month is closed and locked by management, staff cannot retroactively insert or modify invoices or expenses.
4. **"Effortless Store Credit & Prepayments":** Customers with overpayments or returns automatically get store credit that can be applied to future invoices at the click of a button.